Uncovering Characteristics of AI-Rewritten Scientific Writing Using Stylometric and Linguistic Features

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Abstract

With the increasing use of AI in academic writing, identifying stylometric and linguistic features that distinguish AI-generated text from human writing has become an area of interest. This study presents a logistic regression classifier trained on features extracted from PubMed biomedical article bodies, prioritizing interpretability so that the contribution of individual features to the classification decision remains transparent.

1 Introduction

Large language models (LLMs) have become common tools in scientific writing. They are now used to draft, edit, and rewrite academic papers, and a large and growing share of recent publications show signs of LLM involvement. This involvement is usually invisible, and that creates three problems. The first is integrity: when AI use is not disclosed, it becomes unclear who, or what, is accountable for a paper's claims. The second is reliability, because language models can introduce fabricated citations and subtle factual errors that degrade the published literature. The third is trust in peer review, since AI has reached the review process itself, and readers have no dependable way to tell.

Detecting this involvement is difficult. The usual signals are weak: a handful of giveaway words that are trivial to edit out. AI-detection tools have recently appeared, but their reliability is contested, and they act as black boxes that flag text without explaining why. That opacity carries a real cost: false positives wrongly implicate honest authors, and a flag delivered without reasons is hard to contest. To study the problem, we build a paired corpus of human papers and their AI rewrites, and describe each text with a set of interpretable stylometric and linguistic features. We then ask whether these features can detect AI-rewritten scientific writing, and which of them carry the strongest signal. Therefore, this study focuses on answering the following research question: Which extracted stylometric and linguistic features of scientific text are most indicative of it being AI-generated rather than human-written?

2 Research Context

Existing literature has identified several linguistic features that distinguish AI-generated text from human text. Georgiou (2025) showed that when using phonological, morphological, syntactic, and lexical features to distinguish ChatGPT-generated text from human-generated text, phonological features are the most reliable as AI texts tend to favor nasal, alveolar and voiceless consonants. Their study also demonstrated that AI texts use more nouns and coordinating conjunctions, while human texts mostly implement auxiliaries and pronouns.

Joseph et al. (2025) took this methodology further by using stylometric and statistical features to train a Random Forest classifier using outputs generated by GPT-4 and Claude 3, and achieved an F1-score of 0.95.

Demonstrating the increased use of LLMs over time, studies revealed that LLM-associated vocabulary words such as delves, underscores, showcases, potential and crucial have been more heavily used in academic articles from various databases since the implementation of ChatGPT (Kousha and Thelwall, 2026), (Kobak et al., 2025). A 1500% increase in the use of the word ”delve” from 2022 to 2024 was reported.

The present study extends current literature in three ways. First, it includes Gemini-generated text, which was absent from the reviewed literature. Second, it prioritizes interpretability by using logistic regression, whose coefficients allow us to inspect feature importance directly. Third, it incorporates new features such as syntactic parse tree depth, a structural complexity measure not used in any of the reviewed papers. It also combines the use of a substantial amount of extracted features with classification using a logistic regression model, whereas prior work either extracts and plots features or classifies using a smaller feature set.

3 Methodology

3.1 Data Set

Our dataset consists of two parts, one that was written by humans, and one that was generated by Gemini 2.5-Flash. The human part consists of 200 article bodies downloaded randomly from the PubMed API. The main requirement is that the articles must be published between the year 2000 and 2019 to ensure no AI was used to write them. The AI dataset was then obtained by utilizing Gemini 2.5-Flash to rewrite the previously obtained articles using the prompt displayed in Appendix A. We tested several versions of the prompt before selecting the final version. Each article was rewritten paragraph by paragraph to ensure that the final AI-generated article is roughly the same length as the original. This results in a balanced dataset.

Each sample is represented as a feature vector instead of raw text. Given a list of raw text articles we first use a sliding window to extract 5 sentences at a time. The windows overlap by 4 sentences. Each of these 5-sentence windows serves as a single sample. As a result, classification is performed at the five-sentence window level rather than the article level. We then take these 5 sentences and compute the following features for them:

- Average sentence length;
- Average word length;
- Type to token ratio;
- The variation of sentence lengths (burstiness);
- Average parse tree depth;
- Percentage of words that are associated with AI writing according to the literature;
- Percentage of words that we found to be used more in our AI training dataset;
- Function word counts. We looked at the following function words: pronouns, adpositions, determiners, conjunctions, and auxiliary verbs.

Some of these features were specifically chosen because the literature supports them ((Joseph et al., 2025), (Malviya et al., 2025), (Kobak et al., 2025)). Others were chosen to see how much signal they provide for our classification task, compared to the more established ones.

Before training the model on this dataset, we first normalize the training data, to ensure that the coefficients of our logistic regression model are on comparable scales. Finally, the training dataset contains 70% of our articles, and the test set 30%. The split was made in such a way that if an AI-generated article is in the training data, then the original version of this article also belongs to the training data. This ensures there is no data leakage.

3.2 Model

While histogram-based models and XGBoost were tested, ultimately, the model selected for this study was logistic regression as it is an interpretable baseline whose coefficients directly indicate each feature's association with AI-generated text.

3.3 Code

All of the code was written in Python. The main library used is scikit-learn. From this library, we made use of the logistic regression model. The NLTK tokenizer was used to process the raw datasets. Preprocessing for the entire raw data set was kept to a minimum as different features required distinct information. Furthermore, the spaCy library was used to generate parse trees and to find function words using part-of-speech (POS) tagging. Finally we used the google library to interact with Gemini 2.5-Flash.

4 Results

Given that our main goal was to obtain an interpretable logistic regression classifier, we aimed to have a model that can correctly label AI samples as AI and human samples as human.

4.1 Classification Accuracy

As seen in Appendix B Table 2, the logistic regression classifier achieved a macro F1-score of 0.89

on the test set and 0.92 on the training set. Test accuracy was 0.89, with a macro precision of 0.89 and macro recall of 0.89. The confusion matrix found in Figure 1 shows that the model correctly classified 9,490 AI-generated samples/windows and 9,082 human-written samples/windows, with 1,052 AI samples/windows misclassified as human and 1,210 human samples/windows misclassified as AI.

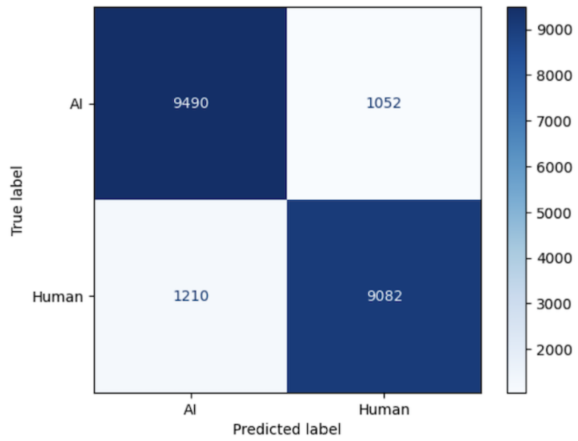

Figure 1: Classifier results on test data

4.2 AI-Word Use

Although existing research such as Kobak et al. (2025) has analyzed increases in specific LLM-associated words after the release of ChatGPT, fewer studies identify which words show the largest usage differences between AI-rewritten and human-written text within a paired corpus. Our analysis of the AI word usage feature reveals that the three words most indicative of AI authorship in our corpus are "authorization", "aligns", and "adhered" as shown in Figure 2.

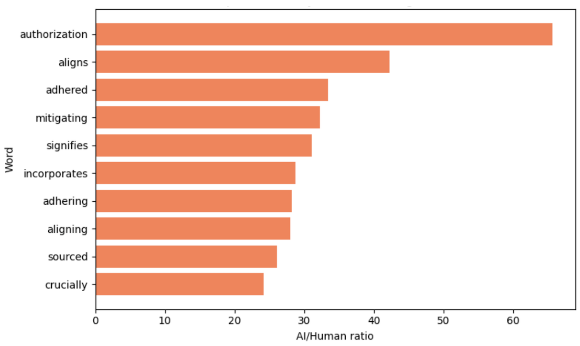

Figure 2: Top 10 words by AI/human usage ratio

4.3 Feature Importance

For feature interpretability, we have outputted the feature coefficients of the logistic regression model, and determined which are most indicative of a paper being AI-generated and of a paper being human-written. Table 1 shows these coefficients, where positive values indicate association with human-written text and negative values indicate association with AI-generated text. Pronoun rate (+0.74), burstiness (+0.70), and auxiliary verb rate (+0.67) are the strongest predictors of human writing, consistent with prior findings that human writing tends to be more varied (Georgiou, 2025). In contrast, learned AI marker rate (-3.82) and fixed AI marker rate (-0.16) are the strongest predictors of AI writing.

Table 1: Feature Coefficients

Feature	Coefficient
Pronoun rate	+0.7367
Burstiness	+0.6975
Auxiliary verb rate	+0.6664
Av Sentence Length	+0.6278
Av Parse Tree Depth	+0.4576
Conjunction rate	+0.3727
Preposition rate	+0.3300
Determiner rate	+0.2777
Av Word Length	+0.0280
Type/Token Ratio	-0.1489
Fixed AI Marker Rate	-0.1592
Learned AI Marker Rate	-3.8243

5 Discussion

The tuned logistic regression model achieves an accuracy and macro F1 score of 0.89, substantially above the approximate random baseline of 0.5. This suggests that interpretable stylometric and linguistic features are effective for detecting AI-rewritten scientific text in our dataset. The macro F1 score of 0.89 on test data and 0.92 on training data suggests a balanced model that does not show strong signs of overfitting.

The confusion matrix shows balanced performance on both classes, without substantially favoring either human-written or AI-generated passages. Given that roughly one in ten passages is misclassified, this suggests that the classes overlap: some AI-written passages resemble human writing according to these stylometric and linguistic features, and vice versa.

Using logistic regression with feature scaling allows for the direct inspection of feature importance through the absolute magnitude of its coefficient, as shown in Table 1. Learned AI marker rate is by far the most important feature, with an absolute coefficient of 3.82, while the next-largest coefficient has an absolute value of 0.74. This suggests that vocabulary is especially informative in this dataset. However, this result should be interpreted carefully. The learned marker vocabulary is specific to our training corpus, Gemini 2.5-Flash, and the rewrite prompt used in this study, so it may not represent universal AI-writing markers.

The remaining coefficients also show that classification is not driven by vocabulary alone. Features such as pronoun rate, burstiness, auxiliary verb rate, and parse tree depth contribute to the decision boundary. Moreover, coefficient importance is conditional on the full feature set: changing or removing one feature can alter the relative importance of others. With the learned AI marker rate only looking at the top 50 words, the hierarchy of feature importance changes (Appendix C, Table 3). For this reason, the coefficients should be interpreted as evidence of how this model separates the two classes, rather than as fixed general properties of all AI-generated scientific writing.

6 Limitations

As any other research, this study has several limitations that should be considered when interpreting the results. First, the dataset is relatively small, consisting of only 200 article pairs, which may affect the generalizability of the classifier. Second, the corpus is drawn exclusively from PubMed biomedical article bodies, meaning the findings may not transfer to other academic areas with different writing styles. Third, while Gemini 2.5-Flash was chosen since it was not prevalent in previous literature, it is a limitation to only rely on one model to generate the AI rewrites since the results may be specific to this setup. Related to this, the rewrite prompt was fixed, and since small changes in the prompt can lead to significantly different outputs, the AI-generated articles may not be representative of AI writing in general. Because the strongest feature is learned from the training corpus, it may partly capture Gemini- and prompt-specific vocabulary rather than universal markers of AI writing. Finally, the study is limited to English-language articles, leaving open the question of whether the same features are discriminative in other languages.

7 Future Research

As mentioned above, our results come from a single model, a single setting, and a single field, so the main task for future work is to test how far they generalize. A natural next step is to rebuild the corpus using rewrites from other language models and prompts, and to check whether the same features still separate human from AI text.

Additionally, the classification model could be evaluated on more complex scenarios. For instance, text generated by a model with no human passage, and real papers that are only partially AI-edited rather than fully rewritten. Testing on these would show whether the detector holds up outside the controlled setting it was built on. Finally, replicating the study across various fields, such as humanities or computer science, would show whether the identified features are general indicators of AI writing or specific to biomedical writing.

8 Conclusion

In this project we trained a logistic regression model to predict if a passage was written by humans, or AI-generated. The model uses stylometric and linguistic features to detect AI writing. The training data consists of 200 articles downloaded from PubMed and their Gemini 2.5-Flash rewrites. The model achieved a macro F1 of 0.89 on the test set, with balanced performance across both classes.

When looking at the model's coefficients we found that the learned AI-marker vocabulary was by far the most informative feature, while pronoun rate, burstiness, and auxiliary verb rate were the strongest indicators of human writing. These findings suggest that AI-rewritten scientific text can be detected using transparent and relatively simple features in our dataset. At the same time our reliance on a single model, prompt, and discipline means the specific feature importances reported here should be read as evidence of how this particular classifier separates the two classes rather than as universal markers of AI writing.

9 Contributions

9.1 Wolter van der Hoef

Wrote the `main.ipynb` notebook, except for some of the markdown text and the feature distribution graphs. Wrote the `text_features.py` skeleton, and the specific features:

- `av_element_length(x)`;
- `av_word_length(x)`;
- `av_parse_tree_depth(x)`;
- `function_word_rates(x)`.

Wrote parts of the `README.md` file. Wrote the Methodology and Conclusion sections of this report.

9.2 Leticia Dupleich Smith

Added commentary and the feature distribution graphs to the `main.ipynb` notebook. Added the following features to the `text_features.py` file:

- `type_token_ratio(x)`;
- `burstiness(x)`.

Wrote parts of the `README.md` file. Wrote the `fetch_articles` Python script to extract PubMed articles and create the human dataset. Wrote the Research Context and Results section of this report.

9.3 Jonathan Cowley

Extended the `fetch_articles` script with article-type filtering, article body recovery, and pagination. Built the AI-rewrite pipeline (`rewrite_papers.py`): paragraph-by-paragraph rewriting with Gemini 2.5-Flash, length matching, the paragraph splitter, and per-paper context caching. Wrote `check_split.py`, a utility to verify the paragraph splitter's output before bulk rewriting runs. Generated the AI side of the dataset, producing the 200 AI-rewritten articles that pair with the human corpus. Wrote the Introduction and Future Research sections of this report.

9.4 Melle Doucet

Added the learned AI marker and fixed AI marker features to `text_features.py`. Implemented these two features into main, tuned the learned AI marker, and wrote the markdown text and the graphs explaining them. Created `model_comparison.ipynb` and its supporting file `model_functions`. Wrote parts of the `README.md` file and the discussion.

References

Georgios P. Georgiou. 2025. Differentiating between human-written and ai-generated texts using automatically extracted linguistic features. *Information*, 16(11):979.

Emmanuel Joseph, Micheal Bennet, and Thomas Kingsley. 2025. Feature-based detection of ai-generated text: An analysis of stylometric and perplexity markers in contemporary large language models. ResearchGate. Preprint.

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Shrikant Malviya, Pablo Arnau-González, Miguel Arevalillo-Herráez, and Stamos Katsigiannis. 2025. Skdu at de-factify 4.0: Natural language features for ai-generated text-detection.

A Gemini Prompt

```
"Rewrite the paragraph below from
the academic paper provided as
cached context.  Restructure
the sentences and reorder the
information; don't just swap
synonyms.  Don't introduce facts
not in the original.  Output
plain text, target ~{target_words}
words.  Paragraph:  {paragraph}
Rewrite:"
```

B Metrics

Table 2: Classification performance metrics.

Metric	Score
Macro F1 (Train)	0.9151
Macro F1 (Test)	0.8914
Accuracy (Test)	0.8914
Accuracy (Train)	0.9153
Macro Precision (Test)	0.8916
Macro Recall (Test)	0.8913

C Additional Feature Insight

Table 3: Feature coefficients (higher is more human-like).

Feature	Coefficient
Burstiness	+0.9035
Auxiliary verb rate	+0.8248
Av Sentence Length	+0.4628
Av Parse Tree Depth	+0.3257
Preposition rate	+0.1667
Pronoun rate	+0.1456
Conjunction rate	+0.1163
Determiner rate	−0.0449
Fixed AI Marker Rate	−0.5654
Av Word Length	−0.6415
Type/Token Ratio	−0.8123
Learned AI Marker Rate	−1.7446